

# ActInf Textbook Group ~ Cohort 3 ~ Meeting 18

## (Chapter 8, part 1)

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all right welcome everyone it's 7 11 23 we're in cohort three in our first discussion of chapter eight so we'll head over to chapter eight and look at some questions and comments but first does anyone wanna bring up any uh just general or thoughts or Reflections on chapter eight um Ollie uh well yes um actually chapter 8 is obviously the title suggests uh is a kind of complementary to uh chapter seven uh in terms of uh being the continuous time counterpart to uh the active inference formulation of discrete time uh so uh as we saw previously in chapter four uh we have both continuous time version and discrete time version of active inference employing different mathematical Technologies and techniques but of course they can be regarded as a kind of uh formulating the uh the same or at least a similar formulation just using uh different mathematical techniques so namely in a discrete time case uh we use uh matrices to represent uh States Transitions and so on um but in a continuous time we use generalized coordinates in order to describe the states and not just I mean discrete matrices for States and also the starting point of formalizing active inference in continuous time also differs a bit because uh in continuous time version we begin with a version or a variant of Itô's stochastic equation by representing a flow plus random variable and then trying to somehow uh formulate the flows in terms of increasingly granular way and yeah tracing the trajectory through the uh State space continuous time State space uh employing that um I mean Taylor expansions and other mathematical Technologies and so on great all of you uh yeah what do you mean by granular here so I mean uh it was a stochastic equation is a very general equation describing any kind of sorry differential stochastic uh phenomenon but in this case we focus on a particular kinds of State space with particular kinds of behaviors so especially in active inference we focus on active agents as opposed to a more General sense of the term uh such as I mean describing the whole behavior of a self-organizing system or even a simple simple system that can be basically described with a simple statistical physics so the focus of active inference as I said is uh the the agents that doesn't that that don't have empty set of active

States right but of course it can also be generalized to other kinds of Agents with empty active States uh but but the active inference framework or fep don't have anything particularly interesting to say about those other kinds of Agents uh that's what I meant of granular we focus on a subset of the systems that can be described through it was a stochastic equation okay cool thank you we cannot come back to some of these uh and also Ali we will be keeping in mind that we need to make the videos for seven eight nine four five sure yeah whatever whatever you want um um okay anyone else want to just have any general thought or question raising their hand or or in the chat on on a on and continuous time I really like that chapters two and three give us the kind of uh two ways to get there to active the low road and the high road the how and the why and now it's like we're in the city and now there's two restaurants or there's like two doors to the restaurant with the discrete and the continuous time so it's another dyad of chapters with seven and eight and another juxtaposition that gives us Parallax and helps us like understand more about each of these two methods by um highlighting on one hand situations where they seem to apply uh more naturally for example the discrete time formulation seems to be a natural fit for discrete decision making tasks like Which slot machine are you going to pull whereas The Continuous time model seems to be a more natural fit for analog perception control systems like proprioception and uh sensory motor Behavior and chapter 8 is also going to get us to hybrid nested models for example a model with two layers where the lower levels continuous time and the higher level is discrete time so that kind of represents like a just a decision-making central nervous system that's dispatching um commands to a continuous time sensory motor body and that is also explored in the sort of um folk psychological active inference work where you can kind of get the the synthesis of continuous and discrete time models anyone else wanna add a comment on eight before we start to look through it yeah I have a question maybe it's nice to verify as my video on let's see um I was wondering if there are any good resources on applications of continuous active inference because um up until now I've worked on two projects with active interns but only working with discrete States state space models and I was wondering if there is any like good papers on continuous active inference and it's it's sort of um representative applications at the endicipation yeah I think this table has some oh yeah yeah I'm so sorry for asking a redundant question no no it's it's it's all good these are not all the applications but let's just look at the topics that they have raised so the bird song is definitely a classic single singing bird and uh to singing birds ocular motor Behavior reflex Behavior more eye Behavior um different cognitive States more eye Behavior this is one of the few papers on mirror neurons um

attention I believe this is attention being modeled in a slightly different approach um this is a continuous time attentional framework whereas there the um Sanfey Smith paper uses um discrete time attention the hybrid and self-organizing but I there's other um citations and and I'm sure you could come up with many more things and also like one of the kind of model annotation tasks that I think would be super helpful in terms of meta-analysis would be like for each paper how could you how can we describe how can we describe the active inference models they used so then we could just like filter a table where are the discrete time models where are the continuous time models where are the nested models where are the models that use yeah yeah yeah thank you for the overview cool so um from the figures figure 4.3 recall that the discrete time and the continuous time generative models have some structural similarities at the same time they have important differences and also partially this is uh signaled with a difference in notation so like hidden states are going to be reflected with  $x$  in continuous time not  $s$  observables are reflected with  $Y$  not  $o$  uh causal interventions and the system are reflected in  $V$  not  $\pi$  and really the most important difference is that whereas in the discrete time you have past present and future explicit state predictions in the continuous time setting you're using the generalized coordinates of motion where  $X$  is the hidden state at a given time and  $X'$  is the derivative of the hidden state how it's changing through time so the  $B$  matrix which is the transition matrix in discrete time describing how states go from one time step to another how the hidden states change here is describing like kind of like the derivative or the integration function because it's describing how derivatives are linked to each other and what causes do in the model is the influence how derivatives change through time here's the Edo calculus type equations that Ali spoke to First we have observable states are a function  $G$  of hidden states that's the flow and some stochastic noise term about  $y$  sensor noise and then  $\dot{x}$  rate of change in  $x$  is a function of the hidden state at the current time and the causal forces on the system at that time plus a fluctuation term on  $x$  action is absent in equation 8.1 this is because action is part of the generative process not the generative foreign are definitions of generative process and generative model the generative model only deals with those variables directly influenced by states external to a Markov blanket what do you think about this Ali uh well yeah actually this is one of the points that that was kind of brought up in in the discussion with Maxwell because as we know generative model I mean if we consider fep as this a kind of General framework or theoretical framework to describe the behavior of the things quote-unquote things that persist through time we can theoretically describe uh the generative model as containing any kinds of different situations both

containing active States and uh or in other words with active States and also with empty set of active States uh so but as I said before I believe active inference is more concerned with the situations that have non-empty set of activist States in other words they have the ability to perform or to undertake or to put to put it in a more uh rigorous terms to kind of engage in a causal intervention with the environment both in terms of updating the generative model and also uh if effectuating some some changes in the environment that can fit better with their generative models so yeah that's I mean what's meant here by directly influenced by States external to a mark of blanket means the states from the environment that can dynamically act upon the markup blanket in order to make it more suitable or more adapted to the environment through the generation of the markup blankets I'm sorry degenerative model and also minimizing the free energy yeah thank you it looks like the difference here between 8.1 and 8.2 other than the Bolding of the equations is like  $V$  a static variable is replaced by a controllable action parameter so now a specific implementation of equations 8.1 8.2 are going to be described so now that uh Ali go for it uh sorry I forgot to mention something uh there was a debate on Twitter a couple of days ago about whether or not fpp can be applied to inert objects such as rocks which is a uh I mean uh one of the debates that's been going on for a long time now but uh you see theoretically it can be applied to any kinds of object even to a manure to rock but the only difference here is that inner truck has an empty set of Acts of the states that's it so if we take a non-empty set of active states such as I don't know biological organisms or sentience agents and so on um that would also employ the same kind of mathematical Technologies but the only difference would be the existence of non-empty set of active States sure I I would almost also say though that it's not that the active states are empty for an inert or a classical or a conservative object that they're just trivial for example just outputting a stationary action state okay so now they're going to give a specific um example of what this pair of equations looks like and there's a point attractor so this is going to be kind of moving us towards a um uh uh continuous time system where there's a fixed attracting point where if the um the actuators below the fixed point it's going to go up and if it's above the fixed point it's going to go down and so this is going to be the kind of model that's used for isocades and for like elbow movements and things like that generative process is specified the hidden state is basically just passed through and the action is basically just passed through  $V$  is in the generative model  $V$   $U$  is in the generative process this connects to the equilibrium points hypothesis again it relates to a proprioceptive prediction about expected proprioceptive feedback coming from for example the limbs or the eyes this enables action to be prediction

fulfillment rather than for example reward maximization or utility maximization directly this relates to an extensive discussion on forwards and inverse modeling and interactionism and instructionism and some of that was explored in the skillful performance live stream now they're going to move from this fixed Point attractor into a spring attractor so now there's going to be an oscillatory uh attraction with dampening and that is going to require not just a uh scalar description of the attractive points but we're going to start moving into this generalized coordinate way of thinking where we're going to be tracking kind of a vector of position and velocity and generalized coordinates you can think of they keep going like you could have put position velocity acceleration and so on so they kind of just work through that and say yeah you could also just write down the generative model where not just the the elbow wants to be at 90 degrees and it's at 80 so it extends out to 90 but now you could have one where it goes and it overshoots 90 and then dampens it back to 90. and you could continue to develop these um uh generalized coordinate based systems or you could think about the fixed points being attractors in a generalized coordinate space this discussion connects us to consideration of dynamical systems more broadly fixed point tractor the spring attractor those were both dynamical systems and so now we're kind of fully in this dynamical systems setting which is natural for continuous time um discussions and um stochastic differential equations and all of this some topics that are certainly relevant for the discrete time setting but are more naturally addressed in continuous time so in chapter five we looked at this spinal cord cross section and it was discussed as a simple uh proprioceptive input descending fixed Point prediction and then the residual was being passed on the error signal was being passed forward to engage bodily movements so like if the proprioceptive data and the descending prediction were identical there's no discrepancy no motion occurs whereas when there is a discrepancy between the expectation of proprioception and the incoming proprioception this triggers um a discrepancy and and um that discrepancy is resolved through movement so now that example from chapter five is being re-approached using the formalisms and the continuous time setting that we see in chapter eight here we have a box on Precision attention and sensory attenuation um this is going to cover several different topics first precision and attention are described as synonymous the that's a pretty General claim however um one can think of as something that's being attended to highly is giving precise inputs and therefore is Salient it's relevant it updates the model a lot if something is not being attended to in the limit that is equivalent to something that has zero Precision it's infinitely ambiguous and therefore it doesn't update the model because it's not Salient it's not relevant so it's kind of a general point about precision and

attention and one way that this plays out in relationship to um motor control and some extremely interesting and unique explanations and predictions arising about Precision attention and motor control in a predictive processing setting is this topic of sensory attenuation in motor generation so attenuation means diminishment and in some ways attenuation even though it sounds like basically the same word is like the opposite of attention it just means it's like dis attention or like uh alleviation of attention um does anyone want to like give a thought or explanation about what is the role of of sensory attenuation in motor initiation and fulfillment okay let's give a thought on it so if one is sitting down in a chair the accurate belief to have is that I am sitting down how without invoking goals or explicit predictions about future time points how do we go from having an accurate belief I am sitting down to having an accurate belief that I am standing up I mean the transition between those two steps basically two things have to happen you have to change your mind thinking you're sitting down to thinking that you're standing up and you have to change the world you actually have to go from sitting down to standing up that's and so how does that happen well the issue is that if you're sitting down and you have a high Precision accurate belief I am sitting down and then you change or you try to change your expectation or your preference or your fixed point or your set point standing up it's going to be like no that's unlikely all of our proprioceptive evidence is telling us that we're sitting down and so um in in that way like the body would kind of be like an anchor and it would pull the cognitive belief back to no actually we're sitting down so if you just try to be like we're sitting down and we know that we're sitting down and you just try to say okay now we're standing up but the body's still sitting down if you have high attention on the proprioceptive inputs then that cognitive belief gets dragged back and so you end up not initiating motor movements and so the resolution of this challenge is you update your belief I am standing up while you're sitting down meanwhile you engage in sensory you release the attention on the and during that alleviation of attention to proprioceptive input you can think about the the Distribution on proprioceptive inputs to actually include standing up so now that is a more viable hypothesis the posterior probability becomes higher then action selection can be engaged to fulfill that expectation about standing up and so voila now you're standing up and you think you're standing up and then you can re-engage Precision so that now you can have that precise belief that you're standing up whereas previously you had a precise belief that you were sitting down and the key step was kind of like you're driving on the road it's like attention is like the gas you pull off the gas make the turn and then re-engage the gas and another example of this is in our isocades when our gaze

is fixed we have high Precision on our visual input when we do an isocate transiently IT suppresses visual precision and that's why during cicades which happens several times a second we don't see like a big blur because it's almost like a blink now it's not your eyelids coming over your eyes but it's actually an attentional blink when you do a cicade which is why the um highly Dynamic input during a cicade is suppressed or not attended to so that we end up having the generative experience of fixed high percentage precision Vision and then we blink into another fixed High Precision visual input with a suppression of attention during the cicade and different pathologies can be understood as challenges associated with initiating movements or with terminating movement or with continuing movements and so different different um pathologies have been associated with this so certain motor dyskinesias where somebody may have a challenge initiating a movement but once they are moving like they walk normally or on the other hand there are um different motor pathologies any thoughts or comments on this this is kind of an interesting topic even just outside of this embodied or or technical setting just like this idea in order for the possible to be actualized there has to be a relaxation on the actual to enable the movements to to a different space and here's the cyclical precision [Music] and this has been Associated heavily with differences in the neurophysiology of vision again just raise your hand or write anything in the uh chat if you want a classic dynamical systems model from ecology is going to be introduced and this is the laka Volterra Dynamics sometimes it's called Predator prey Dynamics or especially in the context of um neural Darwinism and dynamical systems in neuroscience sometimes this is called winnerless competition because as you can see here in the psychological example like it's not that the herbivores are trying to defeat the plants in fact they have a relationship where if the herbivores are transiently succeeding in decreasing the plant numbers then they will have this like demographic collapse and so this is the winnerless competition is you have dynamical oscillations and so this describes ecological Dynamics but also it describes activity um patterns amongst brain regions like two systems that have an oscillatory repressive consequence on each other a repressolator that will have this kind of winnerless competition dynamic um this is just a classic dynamical system and in the SPM textbook the lock of ultra is used to describe these winner lists competitions neural darwinist these three traces plant herbivore and carnivore you can think of them as like in this graph like a plot happening through time or you could imagine a cube with like the X the Y and the Z axis being plant herbivore carnivore Dynamics and so this system can be represented as three traces along the x-axis here or you can imagine it as like a movement in a three-dimensional Cube and so if

we're spending a lot of time up in this corner you can think about what that would mean and so on and then on that Cube just like the girdle Escher Bach book cover you can project from the three-dimensional Cube down to a two Dimension and that would be like putting a camera on one side of the cube and just taking a picture so here we're looking at the herbivore by plant um projection of the cube and it has this kind of oscillatory cycle here it's approximately periodic because there's this dampening but you could design a system that has different behaviors now the lock of Volterra is going to be applied in a few different active inference contexts the first is in the eye blink conditioning and so this is um conditioned stimulus and unconditioned stimulus so this is kind of like a classical scenarian behaviorist paradigm of um of um Behavior action blinking associated with conditioned and unconditioned stimuli using this dynamical systems framework whereas in the discrete time we'd have okay time step one do we get this stimulus or that stimulus or neither and then what do we do and how do we make the decision and then the next time step this happens all of that is happening in this kind of continuous unfolding winnerless competition setting so that's the first example um on the left side that's the blink the unconditioned stimulus um produces a a blink that's alert Association and the right example which um also listened to Thomas parr's bookstream 2.1 when he gives a little more of the history um or we we um we we have a few in the Thomas part 227 page he he he gave some great history here talking about how the generalized filtering and continuous State space approach was early on being the most developed people started wanting to get sequential Dynamics so they developed the kind of model that we're going to see right here like here this is like sequential writing Dynamics clearly there's a repetitive Motif that's being repeated sequentially but then the desire to explicitly sequentially encode beliefs and to engage in explicit planning as inference motivated the work in discrete state-space models where planning as inference is a more natural fit of course by the end of the chapter we're going to be kind of merging these two different kinds of models together but here's an example of lockabletera dynamical systems Dynamics in handwriting setting so it's kind of like a sequence of points that are being targeted and driven through by hand short box 8.2 on learning in continuous models slightly technical note perhaps um sanjeev's textbook will clarify some of these pieces however using LaPlace approximation so a quadratic approximator were able to um use integrals on continuous distributions to specify the attention-weighted evidence grade gradients but this is just kind of little encapsulated box Ali I just wanted to mention a small typo in box 8.2 so the third line in the second block of equations should be minus  $d s \mu \Theta$  so yeah the minus in front  $D$  I suppose is should be



added thank you good dog let me have this one thank you all right now we're going to move into a second class so just like it the first we saw like the simple Point attractor and then the hooke's law the tractor spring now in this dynamical setting we're going to have kind of two levels of sophistication of dynamical systems models the first one was the generalized lockabletera system the second one is the Lorenz system a lorenza tractor which is a canonical example of a of a complex sort of a chaotic system and it's a super fascinating story with Lorenz and his work in weather prediction and that was of course one of the most important moments of the complexity and the computational chaos research of the last hundred years this is also explored a lot more in live stream 34 on um uh Markov blankets and and stochastic chaos I believe where um solitary and coupled lorenza tractors are modeled um it's a super fascinating work um generalized synchrony is going to be unpacked in the setting of Birdsong which was a friston frith paper may have one slightly earlier than this but the 2015 is kind of the classic citation and this is seen as a kind of Exemplar of multi-agent inference problems when they are quote singing from the same hymn sheet as Carl always says and so the bird song attractor is going to leverage the lorenza tractor described above and there's going to be two Lorenz attractors coupled through communication reflecting bird song and some details of those attractors are going to be um described and simulated and the kind of um big takeaway here is that the generalized synchrony is accomplished through communication so here we have the first and the second bird on the X and on the y-axis we can think about this line  $Y = X$  line as being a manifold that would reflect perfect synchrony so if you could describe movement on this two-dimensional plane by appealing only to a movement along a single line you would have reduced a two-dimensional setting into a one-dimensional setting and that's like a dimensional reduction and that's called a statistical manifold before learning while there is overall adherence towards this  $Y = X$  line we see that after learning the manifold um the behaviors tightened immensely around the manifold that represents um the the learning to synchronize between these two birds the two birds joint trajectories can find to an almost one-dimensional Subspace indicating synchronization also it's kind of a cool nod to fractal dimensions like if it was perfectly on this line you'd only need one dimension if it was purely covering the whole 2D space you would need two Dimensions so then you can think about this as being slightly higher than one dimensional this connects back to the earlier attention and and this alternation of listening and singing it's kind of like now this is not all communicative settings and there's more work on Communications but it's kind of a fun Paradigm we're basically like the bird expects there to be singing both Birds expect there to be singing it's like

you know insert your favorite cultural singing moment if you're gonna sing my preferences will be fulfilled through perception and if I'm missing my preferences are going to be filled through action fulfilling my preferences through action is going to reduce my sensory attention to perception until eventually it's like whoa I'm not sure what's happening I'm going to stop singing and then in that silence the other birds gonna go oh my preferences were being fulfilled by listening attentively I'm not hearing it anymore I better reduce this discrepancy by taking action because listening harder isn't going to help because that's not going to make the song happen and so that's singing from the same hymn sheet like Carl says um and it relates to joint uh you know shared shared regimes of expectation and Joint perception action and multi-agent systems and the generalized synchrony doesn't mean lockstep generalized synchrony doesn't mean both birds are seeing identically and then they're both silence generalized synchrony means that there's Mutual information between these two coupled systems such that there is an information geometric manifold that represents their joint movement but it doesn't have to be a lock step same behavior at the same time turn taking is also an example of generalized synchrony 8.5 gets to hybrid discrete and continuous models so here we see a nested model there's a line that's supposed to be between  $\pi$  and  $G$  obviously and in the top level we see the discrete time generative model  $T$  minus  $1/t$  and  $t$  plus one we see all the letters that we know and love from um chapter seven a b d but then within each time step of the higher order model we see the continuous time model with the generalized coordinates of motion and so this was unpacked in live stream 46 octave inference does not contradict folk psychology where the top level was called decision active inference or Dai and the bottom was called motor active inference or m-a-i because somebody can think about the discrete the discrete decision making on the top can be like am I going to raise my left hand or raise my right hand and that's a discreet decision there's no need to um make it an uh um analog but then with the descending command that it's going to be okay raise my right hand then we can engage that whole motor fixed point apparatus of the continuous time model whether we do it with a lockable Terra or a um or lorenza tractor or the hooke's law or just a simple Point attractor so all the kind of continuous time tools that we saw earlier in the chapter we can deploy in the setting of a nested model with a discrete time decision maker on top and a continuous time on the bottom which may articulate better with the seemingly analog nature of perception and action in the last mile this is a little bit of Machinery that helps map between the uh discrete selection of Target locations into the continuous setting where those targets locations can be released and an example of this is using um an isocate where there's a discrete

movement of a Target location followed by a continuous time trajectory of the eye cicating to box 8.3 again raises just some sort of super Curt um analysis of a sophisticated topic which is a gaussian mixture model and like we know the gaussian we love the calcium we know we contractibly optimize it and so on and so on it does an incredible job of approximating distributions that have a Central Mass tendency a lot of statistical tools work incredibly for gaussian but of course not all gaussians are going to describe given complex distributions but the good news whether you approaches from a gaussian mixture model or a hierarchical gaussian filter or just generalized filtering is that it turns out that through composition of gaussians you can build complex distributions that have almost arbitrary characteristics so that's kind of a way to get like the attractability of gaussians but also for example make a bimodal distribution or trimodal distribution this engages the challenge of Bayesian model reduction Bayesian model selection structure learning table 8.1 closes with some discussion of continuous time model advances and that ends the relatively short but pretty dense chapter eight does anyone yeah Olivia yeah uh yeah I have a question I'm not sure I can formulate it uh well but um the thing is you know when we think in terms of discrete time model uh we usually we know we think in terms of regular time steps uh and with this mixed you know uh continuous and discrete time model uh you know it's not possible to to model a behavior in continuous time and you know uh to have um the discrete time step to arrive at different uh I would say you know different diets after different duration within the continuous time model you know when you reach uh you know a Criterion in the continuous time model you get to a Next Step at the top level okay just one thought on that are you referring to a model where the discrete time steps are are actually in homogeneous or yeah yeah yeah yeah yeah so no no no no no no regular in time so for example you will imagine you know a search task uh happening in uh you know in continuous time uh and you reach you know a crit Ion at some point uh and depending on the on the steps you know you can reach that uh earlier or later but when you reach a secretary and you jump you know at the top level uh so you make your decision and you jump at the top level yes so one interesting question is how could that logic be encoded into the generative model another way to view that setting would be you'd have a generative model um that does have like a real um Regular um chronology but then you're modeling samples potentially like poisson distributed samples from a regular continuous or discrete timeline so the delta T of the samples is not necessarily homogeneous but the underlying substrate of the model would be so observations are not coming in regularly actions are potentially not being taken regularly but there is a clock that has regularity that

would be the sort of sampling based approach to dealing with this um time heterogeneity than the other approach would be somehow to figure out through um a composition of discrete and or continuous models um can you encode um a temporal logic when this condition is met then this happens yeah yeah yeah it's an interesting area and I mean this this uh folks psychological or just nested hybrid model I mean this is just a hint and so it would be really awesome I think to see the the Matlab or the other code implementations so we could do okay now let's do this is a two by two Matrix and this one is that um spring attractor okay now this is a three by three Matrix and this one's the lock of Volterra I think that if we could um unroll those systems a little bit more clearly I I think we'd have some really um illustrative Dynamics okay awesome well next week we will um return for a second discussion on chapter eight we'll just go straight to the questions and um Ali I think we'll be ready to do our chapter eight review when the time is right sure yeah all right thank you all uh see you next time